Chapter 3. GROUPS

3.3 Cyclic Groups

3.3.2 The Fundamental Theorem of Finite Cyclic Groups

In this video, we will see a few examples.

Corollary 3.32. *Let $G = \langle g \rangle$ be a cyclic group of order n. Then for any positive integer m we have that $\langle g^m \rangle = \langle g^d \rangle$, where $d = \gcd(m, n)$. Moreover, the order of g^m is n/d.*

Example 3.33. In the (additive) cyclic group $\mathbb{Z}_{30}$, find the order of $\overline{18}$.

Notice that $\mathbb{Z}_{30} = \langle \bar{1} \rangle$ and $\overline{18} = 18 \cdot \bar{1}$ (so, $m = 18$). We have that

$d = \gcd(m, n) = \gcd(18, 30) = 6$.

By Corollary 3.32 we have that the order of $\overline{18}$ in $\mathbb{Z}_{30}$ is $\frac{30}{6} = 5$.

We have that

$$\langle \overline{18} \rangle = \langle \bar{6} \rangle = \{\bar{0}, \bar{6}, \overline{12}, \overline{18}, \overline{24}\}$$

$0 \cdot \bar{6} = \bar{0}$

$1 \cdot \bar{6} = \bar{6}$

$2 \cdot \bar{6} = \overline{2 \cdot 6} = \overline{12}$

$3 \cdot \bar{6} = \overline{3 \cdot 6} = \overline{18}$

$4 \cdot \bar{6} = \overline{24}$

$5 \cdot \bar{6} = \overline{30} = \bar{0}$ in $\mathbb{Z}_{30}$

Example. Consider the cyclic group $G = \langle g \rangle$ with order 20. We want to find the order of g^6.

We begin by computing $\gcd(20,6) = 2$.

So $o(g^6) = \dfrac{20}{2} = 10$

Moreover:

$$\langle g^6 \rangle = \langle g^2 \rangle = \{1, g^2, g^4, g^6, g^8, g^{10}, g^{12}, g^{14}, g^{16}, g^{18}\}$$

$(g^2)^0 = 1$	$(g^2)^3 = g^6$	$(g^2)^6 = g^{12}$
$(g^2)^1 = g^2$	$(g^2)^4 = g^8$	$(g^2)^7 = g^{14}$
$(g^2)^2 = g^4$	$(g^2)^5 = g^{10}$	$(g^2)^8 = g^{16}$

$$(g^2)^9 = g^{18}$$